

Arduino Project Write-Up Format

1. Title

Write the project name clearly.

Example: **Automatic Street Light Using Arduino.**

2. Objective

State what the project is meant to do in one or two lines.

Example: “To automatically turn the street light ON at night and OFF during daytime using an LDR and Arduino.”

3. Components Required

List every part used in the project.

Example:

- Arduino Uno.
- Breadboard.
- LDR.
- LED.
- Resistor.
- Jumper wires.
- USB cable.

4. Circuit Diagram

Draw or insert the circuit diagram here.

Mention the Arduino pins used for each component so the reader can quickly understand the wiring.

5. Code

Paste the complete Arduino sketch here.

Keep the code neat, properly indented, and commented so each major section is easy to follow.

6. Working Principle

Explain how the circuit works in simple steps.

For example, the LDR senses light intensity, Arduino reads the analog value, and the LED turns ON when the light level drops below a set threshold.

7. Explanation of Code

Break the code into parts:

- Variable declaration.
- setup() function.
- loop() function.
- Condition used.
- Output action.

8. Result

Write the final output of the project.

Example: “The street light automatically switched ON in darkness and OFF in bright light.”

9. Applications

Add 3–5 real uses of the project.

Example:

- Smart street lighting.
- Energy saving systems.
- Automatic outdoor lighting.
- Home automation.

10. Advantages

Mention the main benefits.

Example:

- Saves electricity.
- Simple and low cost.
- Easy to build.
- Useful for automation learning.

11. Limitations

Mention what the project cannot do.

Example:

- Works only within sensor range.
- May need calibration.
- Basic version is not weather-proof.

12. Conclusion

End with a short summary of what the project demonstrates.

Example: “This project shows how Arduino can be used for simple automation using sensor input and programmed output.”

Sample Fill-In Template

You can use this exact format:

Title:

[Write project title]

Objective:

[Write project aim]

Components Required:

[List components]

Circuit Diagram:

[Insert diagram]

Code:

[Paste code]

Working Principle:

[Explain how it works]

Code Explanation:

[Explain major parts of the program]

Result:

[Write final output]

Applications:

[Write uses]

Advantages:

[Write benefits]

Limitations:

[Write drawbacks]

Conclusion:

[Write final summary]

Automatic Street Light Using Arduino and LDR

1. Title

Automatic Street Light Using Arduino and LDR.

2. Objective

The objective of this project is to automatically turn the street light ON in darkness and OFF in bright light using an LDR and Arduino.

3. Components Required

- Arduino Uno.
- LDR sensor.
- 10k Ω resistor.
- LED.
- 220 Ω resistor.
- Breadboard.
- Jumper wires.
- USB cable.

4. Circuit Description

Connect one terminal of the LDR to 5V and the other terminal to Arduino analog pin A0. Connect a 10k Ω resistor from A0 to GND to form a voltage divider. Connect the LED anode to digital pin 13 through a 220 Ω resistor, and connect the LED cathode to GND.

5. Working Principle

The LDR changes its resistance according to light intensity. In bright light, its resistance is lower, and in darkness, its resistance increases. Arduino reads the analog value from the LDR, compares it with a threshold, and turns the LED ON when the environment becomes dark.

6. Arduino Code

cpp

```
const int ldrPin = A0;
```

```
const int ledPin = 13;
```

```
int ldrValue = 0;
```

```
int threshold = 500;
```

```
void setup() {
```

```
  pinMode(ledPin, OUTPUT);
```

```
  Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
  ldrValue = analogRead(ldrPin);
```

```
  Serial.print("LDR Value: ");
```

```
  Serial.println(ldrValue);
```

```
if (ldrValue < threshold) {
```

```
  digitalWrite(ledPin, HIGH);
```

```
} else {
```

```
  digitalWrite(ledPin, LOW);
```

```
}
```

```
  delay(500);
```

```
}
```

This code reads the LDR value continuously and switches the LED based on the selected threshold.

7. Code Explanation

- ldrPin stores the analog input pin used for the LDR.
- ledPin stores the digital output pin used for the LED.
- setup() sets the LED as output and starts serial communication.
- loop() reads the sensor value again and again.
- if (ldrValue < threshold) checks whether it is dark.
- If dark, the LED turns ON.
- If bright, the LED turns OFF.

8. Result

The street light automatically turned ON when the light intensity was low and turned OFF when the light intensity was high.

9. Applications

- Automatic street lighting.
- Energy-saving systems.
- Smart city projects.
- Night security lighting.
- Home automation.

10. Advantages

- Simple and low cost.
- Saves electricity.
- Easy to build.
- Useful for automation learning.
- Good beginner project for Arduino.

11. Limitations

- The threshold may need calibration.
- LDR response can vary with environment.
- Basic setup is not weather-proof.
- Best for learning and small-scale demos.

12. Conclusion

This project demonstrates how Arduino can be used with an LDR to create a simple automatic lighting system. It is a useful beginner project for understanding sensor input, decision logic, and output control.

13. Viva Points

- LDR stands for Light Dependent Resistor.
- Arduino reads analog values from the LDR.
- A voltage divider is used with the LDR and resistor.
- The LED turns ON in darkness and OFF in light.
- The project is an example of basic automation.